



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

**Bachelor of Science in Information Technology
First Year - Semester II Examination – November/December 2023**

ICT 1308 –OPERATING SYSTEMS

Time: Three (03) hours

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- This paper contains FIVE (05) questions on THREE (03) pages.
 - Paper counts for 70% of the final evaluation.
 - Answer ALL questions.
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1. a) Define the term “Operating system” in your words and list three (03) examples of operating systems. (05 marks)
 - b) Illustrate the structure of a layered operating system and briefly explain three (03) parts. (05 marks)
 - c) Explain the differences between batch operating systems and multiprogramming operating systems. (05 marks)
 - d) Briefly explain two (02) functions of the operating systems. (05 marks)
 2. a) Differentiate the terms “process” and “programs”. (04 marks)
 - b) List five (05) main process states and briefly explain each of them. (03 marks)
 - c) Illustrate and briefly describe process queues. (03 marks)
 - d) The following table indicates the CPU burst time and arrival time for five processes P0, P1, P2, P3, and P4.

Process	Arrival time	Burst time(ms)
P0	0	10
P1	2	8
P2	4	4
P3	5	6
P4	8	8

- i. Draw the Gantt charts for the following algorithms. (Assume the time quantum for RR is 2ms)
 - i. First Come First Serve (FCFS)-Preemptive scheduling algorithm
 - ii. Shortest Job First (SJF)
 - iii. Round Robin (RR) (05 marks)
 - ii. Find the average turnaround time, waiting time, and completion time for each of the above algorithms. (05 marks)
3. a) Describe the following terms
- i. Process synchronization
 - ii. Race Condition
 - iii. Critical Section (06 marks)
- b) List the primary requirements for process synchronization and briefly explain each of them. (04 marks)
- c) Consider the following solution to the critical section problem involving two processes P_i and P_j . The code is presented below for process P_i .
- ```

Non Critical_Section;
while (turn != i);
Critical_Section
turn = j;
Non Critical_section;

```
- Determine if the solution meets all the required conditions for a fair mutual exclusion. (05 marks)
- d) Compare and contrast the terms “deadlock” and “starvation”. (05 marks)

4. a) There are four strategies for handling a deadlock. Briefly explain deadlock avoidance in your terms. (05 marks)
- b) Briefly explain the advantages and disadvantages of dynamic partitioning in memory management. (05 marks)
- c) Explain the concept of segmentation with the use of an example. (05 marks)
- d) Explain the concept of paging in the Operating System. (05 marks)
5. a) List down three (03) main file access mechanisms and briefly explain one of them. (05 marks)
- b) Briefly describe the six steps of file system structure with the aid of a diagram. (05 marks)
- c) Explain how the CPU communicates with control registers and device data buffers. (05 marks)
- d) Explain the difference between precise and imprecise interrupts. (05 marks)

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